
DAME-BS:

*Distribution Aware Mean Estimation via Binary Search under
user-level Local Differential Privacy*

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Motivation

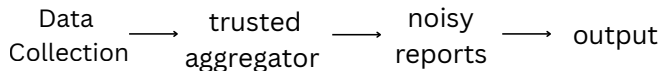
- ▶ Modern ML and statistics require large datasets from individuals.
- ▶ Rising concerns about user privacy, ethics, and legal frameworks.
- ▶ Need algorithms that protect users while retaining statistical utility.

Differential Privacy

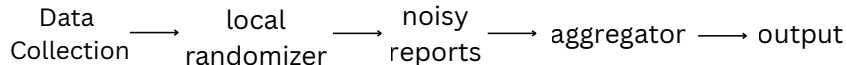
- ▶ Differential Privacy provides a formal mathematical framework to define privacy.
- ▶ **Central DP:** Trusted curator collects raw data, adds noise after aggregation.
- ▶ **Local DP (LDP):** Each user randomizes their data locally before sending it and no trusted party required.

Central vs Local DP

Central Differential Privacy



Local Differential Privacy



Item-level vs User-level DP

- ▶ **Item-level:** each user contributes a single sample.
- ▶ **User-level:** each user may contribute multiple samples (e.g., time series, many ratings).
- ▶ User-level privacy should protect all of a user's samples as a group.

Goal

Problem: Mean estimation under user-level LDP (univariate and multivariate, ℓ_∞ support for multivariate).

Two obvious private-mean strategies:

1. Privatize each raw sample, then average.
2. Average locally (per user) then privatize the per-user mean.

Why these can fail ?

1. **Privatize-each-sample:** To meet a total user-level budget ε you must split privacy across m events. If each event gets $\varepsilon_{\text{evt}} = \varepsilon/m$, Laplace noise per sample has scale $\propto \text{range}/\varepsilon_{\text{evt}} = m \cdot \text{range}/\varepsilon$. Averaging the m noisy samples leaves variance that scales like $\sim m \cdot (\text{range}^2/\varepsilon^2)$ i.e. it can *grow* with m .
2. **Average-then-privatize (user-level):** Privacy must protect replacement of all m samples of a user, so the sensitivity of the per-user mean is still $\sim \text{range}$. The Laplace scale is $b = \text{range}/\varepsilon$, which does not decrease with m . Thus simply averaging locally *does not* reduce required noise under user-level LDP.

Our Focus and Approach

- ▶ **Problem:** Mean estimation under user-level LDP (univariate and multivariate, ℓ_∞ support for multivariate).
- ▶ **Approach:** We take inspiration from Kent et al., 2024 and Pla et al., 2024 and split estimation into two stages:
 1. **Localization:** Find a small interval that likely contains the true mean using a noisy binary search routine.
 2. **Estimation:** Run a refinement estimator inside that localized region.

Our Contributions

- ▶ **Noisy binary-search localization:** A subroutine `Attempting_Insertion_Using_Binary_Search` that returns a high probability interval containing the true mean. Faster in practice and designed for the user-level LDP setting (equal samples per user).
- ▶ **Empirical evaluation:** Practical comparisons (on synthetic and real datasets) with the algorithms proposed by Kent et al., [2024](#) and Girgis et al., [2022](#).
- ▶ **Theory:** Non-asymptotic upper bounds on expected squared ℓ_2 error for our proposed estimator for both univariate and multivariate case.

Notations

- ▶ **Users:** $[n] = 1, \dots, n$. Each user u has m i.i.d. samples $X_{1:m}^{(u)} \sim \mu$, supported on $[-1, 1]$.
- ▶ **Data:** Each $X^{(u)}$ is mapped to private view $Z^{(u)}$ defined on $\mathcal{Z}$ via random mapping Q and $\sigma(\mathcal{Z})$ denotes σ -algebra on $\mathcal{Z}$.
- ▶ **Privacy Parameter:** We fix $\alpha \geq 0$.
- ▶ **User empirical mean:** $\bar{X}^{(u)} = (1/m) \sum_{i=1}^m X_i^{(u)}$.
- ▶ **Goal:** Minimize $\mathbb{E}[\|\hat{\theta} - \theta\|_2^2]$ where $\theta = \mathbb{E}_\mu[X]$ is true mean and $\hat{\theta}$ is the estimator we get using privatized reports.
- ▶ **Notation reminders:** $a \vee b = \max\{a, b\}$, $a \wedge b = \min\{a, b\}$, $O(\cdot)$ for limiting behavior.

Local Differential Privacy

Definition : Given data $\{X^{(u)}\}_{u=1}^n \subset \mathcal{X}$, consider a family of conditional distributions $\{Q_i\}_{i=1}^n$ where $Q_1 : \sigma(\mathcal{Z}) \times \mathcal{X} \rightarrow [0, 1]$ and $Q_i : \sigma(\mathcal{Z}) \times \mathcal{X} \times \mathcal{Z}^{i-1} \rightarrow [0, 1]$ for $i \in \{2, \dots, n\}$. We say that such a collection of conditional distributions $\{Q_i\}_{i=1}^n$ satisfies α -LDP if, for all $i \in \{1, \dots, n\}$

$$\sup_{S \in \sigma(\mathcal{Z})} \frac{Q_i(Z^{(i)} \in S \mid X_{(1:m)}^{(i)} = x_{(1:m)}^{(i)}, Z^{(1:i-1)} = z^{(1:i-1)})}{Q_i(Z^{(i)} \in S \mid X_{(1:m)}^{(i)} = x'_{(1:m)}{}^{(i)}, Z^{(1:i-1)} = z^{(1:i-1)})} \leq e^\alpha,$$

$$\forall X^{(1:n)}, X'^{(1:n)} \in \mathcal{X}^n, Z^{(1:i-1)} \in \mathcal{Z}^{(1:i-1)}$$

- ▶ Two-phase strategy:
 1. a *localization* phase that rapidly narrows the range in which the population mean θ can lie, and
 2. an *estimation* phase that privately averages user-level statistics after projecting them into the localized interval.
- ▶ Localization reduces sensitivity (hence required noise) and improves final MSE given a fixed privacy budget.

Localization Phase I

Algorithm Attempting_Insertion_Using_Binary_Search (α, δ, n, m)

- 1: **Initialize:** $b_{\max} = \max \left\{ \frac{2n \ln(3/2)}{\ln\left(\frac{8m}{9 \ln(12)}\right)}, \frac{2}{(2\pi_\alpha - 1)^2} \ln(2n/\delta) \right\}$ and $t_{\max} = \lfloor \frac{n}{b_{\max}} \rfloor$
- 2: Divide n users into $t_{\max}$ set of users (say Z_t), each of size $b_{\max}$ and set $L = -1$ and $R = 1$
- 3: **for** $t \leftarrow 1$ **to** $t^{\max}$ **do**
- 4: Divide $[L, R]$ into three equal-length intervals. Let $\gamma_t = \frac{L-R}{3}$.

$$I_1 = [L, L + \gamma_t], \quad I_2 = [L + \gamma_t, R - \gamma_t], \quad I_3 = [R - \gamma_t, R]$$

Localization Phase II

Algorithm Attempting_Insertion_Using_Binary_Search (α, δ, n, m)

5: **for** $u \leftarrow 1$ to b_{max} **do**

$$V_{l_1,t}^{(u)} = 1_{\{\bar{X}^{(u)} \in l_1\}} \quad \tilde{V}_{l_1,t}^{(u)} = \begin{cases} V_{l_1,t}^{(u)}, & \text{w.p. } \frac{e^\alpha}{1+e^\alpha} \\ 1 - V_{l_1,t}^{(u)}, & \text{otherwise} \end{cases}$$

and

$$V_{l_3,t}^{(u)} = 1_{\{\bar{X}^{(u)} \in l_3\}} \quad \tilde{V}_{l_3,t}^{(u)} = \begin{cases} V_{l_3,t}^{(u)}, & \text{w.p. } \frac{e^\alpha}{1+e^\alpha} \\ 1 - V_{l_3,t}^{(u)}, & \text{otherwise} \end{cases}$$

where $\bar{X}^{(u)} = \frac{1}{m} \sum_i X_i^{(u)}$ i.e. empirical mean of m available samples of user u .

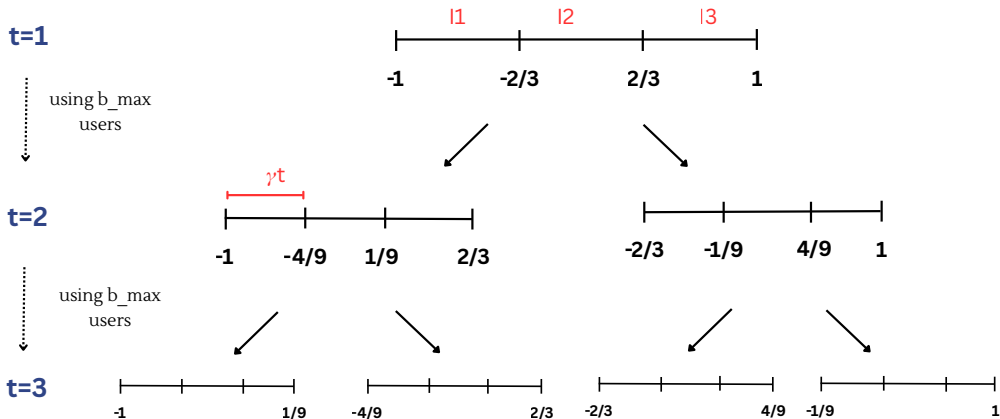
6: **end for**

Localization Phase III

Algorithm Attempting_Insertion_Using_Binary_Search (α, δ, n, m)

7: $\mathcal{V}_{l_1,t} = \sum_{u=1}^{b_{max}} \tilde{V}_{l_1,t}^{(u)}$ and $\mathcal{V}_{l_3,t} = \sum_{u=1}^{b_{max}} \tilde{V}_{l_3,t}^{(u)}$
8: Remove the interval $l_{\hat{j}}$ where $\hat{j} = \arg \min_{z \in \{1,3\}} \mathcal{V}_{l_z,t}$
9: **if** $\hat{j} = 1$ **then** $L = L + \gamma_t$
10: **else** $R = R - \gamma_t$
11: **end if**
12: **end for**
13: return the interval $[L, R]$

Localization Phase:



Estimation Phase I

- ▶ **Input:** Localized Interval $I = [L, R]$ from localization phase.
- ▶ Expand to

$$\mathcal{I} = [\tilde{L}, \tilde{R}] = [L - \delta', R + \delta']$$

where $\delta' = \sqrt{\frac{1}{m} W_0 \left(\frac{32\alpha^2 nm}{81} \right)}$ and $W_0(\cdot)$ denotes the Lambert Function.

- ▶ Project means of remaining half users $\bar{X}^{(u)}$ onto $\mathcal{I}$.
- ▶ Add Laplace noise and compute $\hat{\theta}^{(u)} = \Pi_{\mathcal{I}} (\bar{X}^{(u)}) + (\text{scale}) l_u$

Estimation Phase II

- Average privatized projected means and calculate final estimator

$$\hat{\theta} = \frac{2}{n} \sum_{u=\frac{n}{2}+1}^n \hat{\theta}^{(u)}$$

.

- Noise scale used in Laplace mechanism:

$$scale = \frac{2}{\alpha} \max \left\{ \left(\frac{2}{3} \right)^{\left\lfloor \frac{\ln(\sqrt{(9 \ln(12))/8m})}{\ln(2/3)} \right\rfloor}, \left(\frac{2}{3} \right)^{\left\lfloor \frac{n(2\pi\alpha-1)^2}{2 \ln(2n/\delta)} \right\rfloor} \right\} + \frac{2\delta'}{\alpha}$$

Univariate Mean Estimation

- ▶ **Inputs:** n, m, α , user samples.
- ▶ **Step 1 (localization):** Use first $n/2$ users to perform `Attempting_Insertion_Using_Binary_Search` and get $I = [L, R]$.
- ▶ **Step 2 (expand):** Compute δ' and expand interval I to get $\mathcal{I} = [\tilde{L}, \tilde{R}] = [L - \delta', R + \delta']$.
- ▶ **Step 3 (estimation):** Use remaining $n/2$ users and project $\tilde{X}^{(u)}$ onto $\mathcal{I}$, add *Laplace(scale)* noise and compute send $\hat{\theta}^{(u)}$.
- ▶ **Step 4 (aggregate):** Output

$$\hat{\theta} = \frac{2}{n} \sum_{u=\frac{n}{2}+1}^n \hat{\theta}^{(u)}$$

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Multivariate Mean Estimation

- ▶ For multivariate mean estimation, we consider the following family of distribution:

$$\mathcal{P}_d = \{P : \text{supp}(P) \subseteq \mathbb{B}_\infty(1) \subset \mathbb{R}^d\}$$

with the mean of a distribution P denoted as $\theta(P) = \mathbb{E}_P[X]$

- ▶ Estimate each coordinate independently with univariate procedure. Partition users into d disjoint groups. For coordinate j , perform DAME_with_binary_search with n/d users and compute $\hat{\theta}_j$
- ▶ Assemble and get the final vector as $\hat{\theta} = \sum_{j=1}^d \mathbf{e}_j \hat{\theta}_j \in \mathbb{R}^d$ where $\mathbf{e}_j$ is the j^{th} standard basis vector of $\mathbb{R}^d$

Assumptions:

In order to find an upper bound on the expected squared error, we assumed the following:

1. (Bounded support of μ) We assume that all the samples of users are i.i.d sampled from the distribution μ and support of μ lies in $[-1, 1]$.
2. (High Privacy Regime) We assume that the privacy parameter $\alpha \in (0, 1]$ throughout.
3. (Enough Samples per user) The number of samples per user should be atleast 7 i.e. $m \geq 7$.

All of the above assumptions are very common and have been considered in previous works before (Duchi et al., 2018; Bassily and Sun, 2023; Kent et al., 2024; Pla et al., 2024).

Localization Guarantee

Lemma: If the previous assumptions and the following inequality on δ (probability of failure):

$$\delta \geq 2ne^{-\frac{n(2\pi_\alpha-1)^2}{2}}$$

are satisfied, then the interval $[L, R]$ returned by the algorithm

`Attempting_Insertion_Using_Binary_Search` contains the true mean with probability at least $1 - \delta$ and the following equality is satisfied if no round of the algorithm fails:

$$R - L = 2 \max \left\{ \left(\frac{2}{3} \right)^{\left\lfloor \frac{\ln(\sqrt{(9 \ln(12))/8m})}{\ln(2/3)} \right\rfloor}, \left(\frac{2}{3} \right)^{\left\lfloor \frac{n(2\pi_\alpha-1)^2}{2 \ln(2n/\delta)} \right\rfloor} \right\}$$

Univariate bound

Theorem: We assume that all the assumptions mentioned previously are satisfied. Then, the following inequality will hold for the algorithm DAME_with_binary_search for $\alpha \in (0, 1]$:

$$\mathbb{E}[(\hat{\theta} - \theta)^2] \leq \frac{389}{nm\alpha^2} \{\ln(2nm\alpha^2) \vee 1\} + 16 \max \left\{ 2n \exp \left(\frac{-n(2\pi_\alpha - 1)^2}{2} \right), \right. \\ \left. \frac{9}{\alpha\sqrt{8}} \exp \left(-\frac{1}{4}(2\pi_\alpha - 1)\sqrt{n} \right) \right\}$$

Multivariate bound

Theorem: If $\text{supp}(P) \subseteq \mathbb{B}_\infty(1)$ and all assumptions mentioned earlier are satisfied, the following inequality will hold:

$$\mathbb{E}[\|\hat{\theta} - \theta\|_2^2] \leq \frac{389d^2}{nm\alpha^2} \left\{ \ln(2(n/d)m\alpha^2) \vee 1 \right\} + 16 \max \left\{ 2n \exp \left(\frac{-n(2\pi_\alpha - 1)^2}{2d} \right), \right. \\ \left. \frac{9d}{\alpha\sqrt{8}} \exp \left(-\frac{1}{4}(2\pi_\alpha - 1)\sqrt{\frac{n}{d}} \right) \right\}$$

Experiments: Overview

- ▶ We evaluated DAME-BS on both synthetic and real-world datasets and compared it with Kent et al., [2024](#) and Girgis et al., [2022](#).
- ▶ **Experiments include:** univariate and multivariate mean estimation, 50 trials per configuration (30 for multivariate), varying α (privacy), n (number of users), m (samples per user), and d (dimension).

Synthetic Datasets

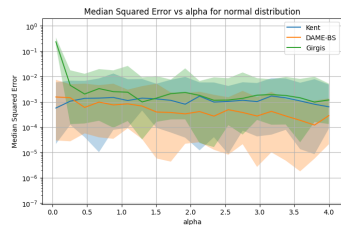
For synthetic data we used four distributions with different properties to ensure a comprehensive evaluation:

- ▶ Gaussian: $\mathcal{N}(\theta, 1)$,
- ▶ Uniform: $\mathcal{U}([\theta - 1, \theta + 1])$,
- ▶ Student- t with $\text{df} = 3$,
- ▶ Binomial with number of trials 50 (discrete setting).

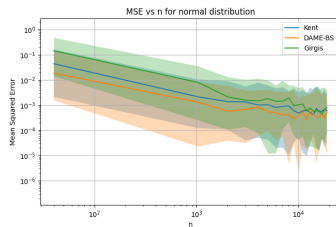
For multivariate experiments we generated independent coordinates with the same marginal distributions. After generation we linearly rescaled values to the range $[-1, 1]$.

Results on Synthetic Datasets

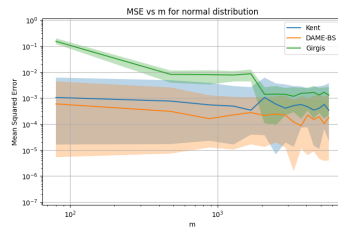
Univariate case



(a) Median squared error vs α



(b) Median squared error vs n

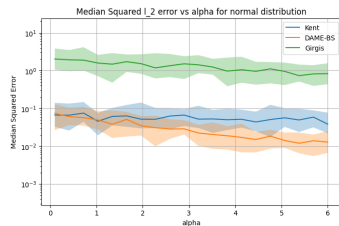


(c) Median squared error vs m

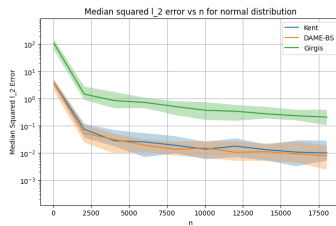
Figure: Median squared error across trials with 10th–90th percentile bands (shaded)

Results on Synthetic Datasets

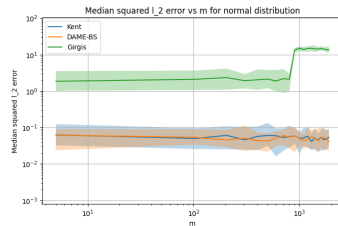
Multivariate case



Median squared error vs α



Median squared error vs n



Median squared error vs m

Figure: Median squared ℓ_2 error across trials with 10th–90th percentile bands (shaded).

Real World Datasets

1. **U.S. equity returns (NASDAQ)**: daily closing prices for $n = 1,584$ stocks using `y_finance`. private mean estimation on log-returns with $m = 249$ data points per stock.
2. **MIMIC-III (medical)**: heart-rate time series for patients ($n = 48$ patients with $m = 11$ samples each).
3. **GLOBEM Fitbit**: wearable and phone-derived measures (location, phone usage, activity, sleep). We focused on step counts across four daily segments and total sleep duration per day of $n = 40$ users with $m = 88$ samples each.

Preprocessing and scaling:

- ▶ We rescaled all data to $[-1, 1]$ using global min-max scaling across retained samples.
- ▶ To ensure equal m per user, we discarded users with fewer than the threshold samples and dropped missing values. Per-user samples were shuffled before truncation to avoid ordering effects. We leave the case for heterogeneous m for future work.

Results on Real World Datasets

	MIMIC (Heart Rate)	GLOBEM (sleep)	GLOBEM (steps)	NASDAQ (log returns)
n	48	40	40	1584
m	11	15	88	249
Scaled Error (<i>DAME-BS</i>)	0.0194 <i>(0.00029 - 0.11350)</i>	0.0327 <i>(0.00136 - 0.16312)</i>	0.3842 <i>(0.12570 - 0.73186)</i>	0.0015 <i>(0.00011 - 0.00558)</i>
Scaled Error (<i>Kent</i>)	0.0242 <i>(0.00055 - 0.11895)</i>	0.0423 <i>(0.00144 - 0.20973)</i>	0.4985 <i>(0.18111 - 0.91402)</i>	0.0050 <i>(0.00106 - 0.01363)</i>
Scaled Error (<i>Girgis</i>)	0.1028 <i>(0.00537 - 0.48552)</i>	0.1483 <i>(0.00886 - 0.59607)</i>	0.0353 <i>(0.00100 - 1.92486)</i>	0.0246 <i>(0.01758 - 0.03297)</i>

Theoretical Comparisons

- ▶ Upper bound is asymptotically optimal up to log factors for the univariate case.
- ▶ DAME-BS, Kent, and Girgis share the same rate with rate $O(1/nm\alpha^2)$ in the univariate case.
- ▶ Differences appear in constants, remainder terms, and dimension dependence.

Theoretical comparisons

- ▶ Univariate (discarding log-factors):
 - ▶ item level ($m = 1$): $O((n\alpha^2)^{-1})$ Duchi et al., 2018
 - ▶ user level ($m \geq 1$): $O((nm\alpha^2)^{-1})$ Kent et al., 2024; Girgis et al., 2022 and DAME-BS
- ▶ Multivariate (ℓ_∞ support):
 - ▶ DAME-BS and Kent: $O((d^2/nm\alpha^2))$
 - ▶ Girgis (with Hadamard-style transform): $O((d/nm\alpha^2))$ up to logs. It has better dimension dependence asymptotically.

Future Work

- ▶ **Introducing Backtracking** : Note that Remainder term in our bound decays in n as $O(e^{-\sqrt{n}})$ (slower than other methods which is $O(e^{-n})$). It is due to “*All-or-nothing*” binary decisions. We can mistakenly prune the correct interval. We want to revisit pruned intervals when decisions are uncertain to reduce failure probability and tighten high probability bounds.
- ▶ **Heterogenous m** : Extend algorithm and theory to varying per user sample counts (distribution-aware batching).
- ▶ **Better multivariate bounds**: Extend the algorithm and analysis to the case where the data support in the ℓ_2 unit ball and have better dimensional dependent theoretical bound.
- ▶ **Broader applications**: Plug the localization primitive into applications like sparse linear regression, private optimization, and nonparametric density estimation.

Conclusion

- ▶ Introduced DAME-BS, an adaptive noisy binary-search localization + private estimation pipeline for user-level LDP mean estimation.
- ▶ Achieves the same dominant asymptotic rate as recent baselines (up to logs) for univariate case while being computationally lightweight and empirically competitive or superior across many regimes.
- ▶ Next steps include make localization more robust (backtracking), handle heterogeneous m , and tighten theoretical guarantees while keeping the method practical.

References I

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- [5] Corentin Pla, Hugo Richard, and Maxime Vono. *Distribution-Aware Mean Estimation under User-level Local Differential Privacy*. 2024. arXiv: [2410.09506](https://arxiv.org/abs/2410.09506) [stat.ME]. URL: <https://arxiv.org/abs/2410.09506>.